

sources (mostly press releases), all laying claim to have developed or used AI in some form or the other. And what were the topics covered? Everything from smart keyboards (predictive text, essentially) to digital assistant devices (Google Home and Amazon Echo types), and also from stock investment bots to e-commerce suggestion bots, it was all there.

Remember, this is the stuff that gets through my very strict spam filter, and are from people whom I have specifically chosen to accept emails from. Honestly, it's like *everything* is AI these days. Or maybe *nothing* is AI these days.

Definitions

So what is really the accepted definition of AI? The dictionary would have you believe that AI is "simulated intelligent behaviour". Basically, whether or not a machine can imitate intelligent human behaviour. Thus, if we go by the dictionary, it's pretty much all of the stuff that people claim is AI these days. Would you consider being able to read and then transcribe words from a page to be something only humans with intelligence can do? I ask, because OCR software have been doing it for some time now. Are those AI? What about translating the written word? Got something written down in Hindi and want it translated into English? Assuming you don't speak one of the languages, you'd something like Google's free translator service. So is that AI then?

Many people are not happy with such a vague definition of AI, and neither are we at Digit happy to set the bar so low. So we need another definition...

Turing Test

This is the point where it becomes obligatory to mention the Turing test. When Alan Turing came up with it back in 1950, it seemed foolproof and very futuristic. In fact, there were surely those who scoffed at him and wondered what on earth he'd been smoking! When calculating

machines were the order of the day, imagining a machine could ever pass the Turing test was nearly impossible. In case you've been living under a rock, the Turing test is basically a setup where a human (A) and a computer (B) are behind a curtain, and an examiner (C) is on the other side, writing out questions to both, and then receiving back written (printed out) answers from both A and B. C is tasked with trying to figure out whether A or B is the computer, based solely on the answers he/she receives. If C is unable to distin-



Alan Turing. Genius. Period.

guish between both, or if C picks the wrong answer and identifies A as the computer, then the computer is said to have passed the Turing test and is considered to be an AI.

To be fair, Turing was an unparalleled genius in the field of computers, and even back in 1950, much before the silicon era, he knew better than to focus on machines developing human intelligence. His test was essentially a take on a popular parlour game called the imitation game.

Turing never claimed that a machine that passed his test would be 'intelligent', he merely stated that it would be able to 'mimic intelligence'. He never believed that a machine could gain consciousness.

Lighthill Report

Of course, this didn't stop people

from claiming that their machines had passed the Turing test and were therefore intelligent. In the 70s, many scientists were working on AI, and all claimed to be on the verge of a breakthrough, while others were extremely sceptical. In order to get to the bottom of the matter, and try and solve the debate once and for all, the British Science Research Council asked James Lighthill, a well-renowned British scientist at the time, to undertake a study and file a report on AI. In 1973 he completed and published his report titled Artificial Intelligence: A General Survey, in which he was very, very critical of the accomplishments of AI research. Although he did praise the work done in very specific and niche fields such as those for trying to solve small and specific problems, he was critical of all general purpose AI research.

The press caught on, and eventually turned public opinion against money being wasted on a pipe dream. The British government cut off funding for all AI research, and it started what would be later called the AI Winter (like Nuclear Winter). Even the Americans undertook a similar study after hearing about Lighthill's report, and the US government also massively cut back on AI funding.

Since then, the field of AI (much like a lot of other things) has undergone a bubble phase where it's super hyped, only to then have the bubble pop and re-enter AI Winter. A lot of this yo-yoing is caused by people like us, the press, because it's the easiest thing for journalists to get carried away with a fad or a new hype, only to then quickly realise that they were wrong, and write scathing takedowns.

So what is AI?

Coming back to the original question we asked, the answer is, sadly, we still don't know.

Some would be happy with the dictionary definition (not us), while most would be happy if it passed the Turing Test (again, not us). We cer-